

7.1 Integration by Parts - continued

Integration by Parts

$$\int f(x)g'(x) dx = f(x)g(x) - \int g(x)f'(x) dx$$

Let $u = f(x)$ and $v = g(x)$. Then, $du = f'(x) dx$, $dv = g'(x) dx$ and we can rewrite our formula as

$$\int u dv = uv - \int v du$$

There are occasions where we must use the substitution method before using integration by parts.

Example 6

$$\int e^{\sqrt{x}} dx$$

Sometimes we need to manipulate the integrand before performing integration by parts.

Extra Example

$$\int x^5 \sqrt{x^3 + 1} \, dx$$

Integration by Parts Formula for Definite Integrals

Example 7

$$\int_0^{\pi} x \cos\left(\frac{x}{4}\right) dx$$

Example 8

$$\int_0^1 \tan^{-1}(x) \, dx$$

Example 9

Do not integrate the following functions. Determine whether we should use simplification, the substitution method, or integration by parts for the following:

1. $\int x^3 e^x \, dx$

2. $\int (x+2)(x^3+1) \, dx$

3. $\int (x+1)\sqrt{x+1} \, dx$

4. $\int x^2 e^{x^3} \, dx$

5. $\int x \cdot \frac{x^3+1}{x^2} \, dx$

6. $\int x \cdot \frac{x}{x^3+1} \, dx$

Example 10

What is the minimum number of times you must perform integration by parts to integrate the following?

1. $\int x^5 e^x \, dx$

2. $\int e^{2x} \cos(5x) \, dx$

3. $\int (x+1)(x^2-1) \, dx$

4. $\int x^2 \ln(x) \, dx$